

3D RECONSTRUCTION & MODELING-SIMULATION OF THE DEVELOPMENT AND GROWTH OF A PLANT BRANCH

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Abstract

The objective of this master thesis was the development of a topological model simulating the architecture of a crop. As a case study, we have taken a branch of an orange tree.

For the establishment of this study, some necessary experimental measurements characterizing the architecture of the branches was considered (length, thickness and base top, number and position of internodes, positions and angles of the various ramifications, etc.).

To convert these measurements into a script for 3D visualization in 3DSMAX, a software has been developed in WinDev.

The second part of this work was to test the L-system approach as a means of characterizing the architecture and plant growth. This approach resides in describing the plant in a formal way. To apply this approach, a converter which processes and converts the measurements in formal language and an algorithm that compiles OL-system have been developed for the 3D reconstruction.

This approach has been subsequently be supplemented by the development of a stochastic L-system that is based on the probability distribution combined with a Markov process enabling the simulation of the development of the branch architecture based on analysis of the current architecture.

Then a growth module (length, diameter, etc ...) was incorporated to realize the synchronism between development and growth.

Finally, a forecast of the orange branch development and growth at a later stage was designed in 3D and compared with the initial state.

Conclusions

The purpose of this Master's work was to develop topological models simulating the architecture of a plant branch. As an example of a branch, we took the case of that of the orange tree.

As a basis for study, an experimental work consisted in taking the necessary measures (length, base and top thicknesses, number and position of internodes, positions and angles of the different branches, ect ...), characterizing the architecture of the branches. Have been realised. From these measurements, a software was developed to convert these measurements into a script for 3D visualization.

The bibliographic synthesis has highlighted the importance of L-systems as an approach to characterize the architecture and growth of plants. This approach consists of describing the plant in a formal way. To apply this approach, a converter that processes and converts measurements made in formal language as well as an algorithm that compiles a 0L-system, L-stochastic system have been developed for 3D visualization.

Indeed, the data of the samples, after being taken from an Excel database, have undergone conversion processing to a formal language including all the geometric information (orientations and measures) and hierarchical information (rank and degree of insertion). These formal sample languages required analysis to extract the keys to evolution and the hidden process.

Two analyzes were made and developed as an application under Windev;

The first is a hierarchical analysis and is based on the Markov process, behavior modeling (growth rule: exponential evolution, linear evolution, phenomenon of delay, death) of the organs (internode) was made in order to identify temporary series and finally extract a transition matrix helping to simulate the hierarchical prediction, in addition to a proposal for an adjustment of the matrix using the Fibonacci sequence.

The second is a geometric analysis based on the sigmoid function, on the normal law and on the angle of phyllotaxis.

After these different analyzes, input parameters (Axiom, Matrix, Geometric Evolution Function) were used to predict a future state of the branches using a plugin MAXscript developed under 3DSMax where the stochastic L-system and the Markov process are combined. The predicted growth and 3D development results are acceptable with a correction of the transition matrix.

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Notes and references

In order to be able to make predictions of future growth and development finer, the algorithm could be improved by relying on fast methods of simulation and on an adjustment of the future random choice knowing the antecedent states. Indeed, any estimate made could be improved by an even more complex set of conditions (conditioning Fibonacci ...) and a speed of simulation. On the other hand, the collection of experimental data can possibly be facilitated with approaches of image processing and epipolar geometry. [45] [46] [47] [48]. Also, the integration of the substructure method allows a fast simulation. [49] [50]. All these aspects offer new perspectives for the continuation of this study.

As you knew that the Textile is dependent on the raw material, and this one is more often of vegetable origin, we can develop more models that are sophisticated and which make it possible to improve the production of the cotton for example.

The beginning always begins with experience and observation, and this thesis approaches the mechanism of vegetal growth.

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